VIT®

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

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COURSE NAME: AGILE DEVELOPMENT PROCESS AND DEVOPS LAB.

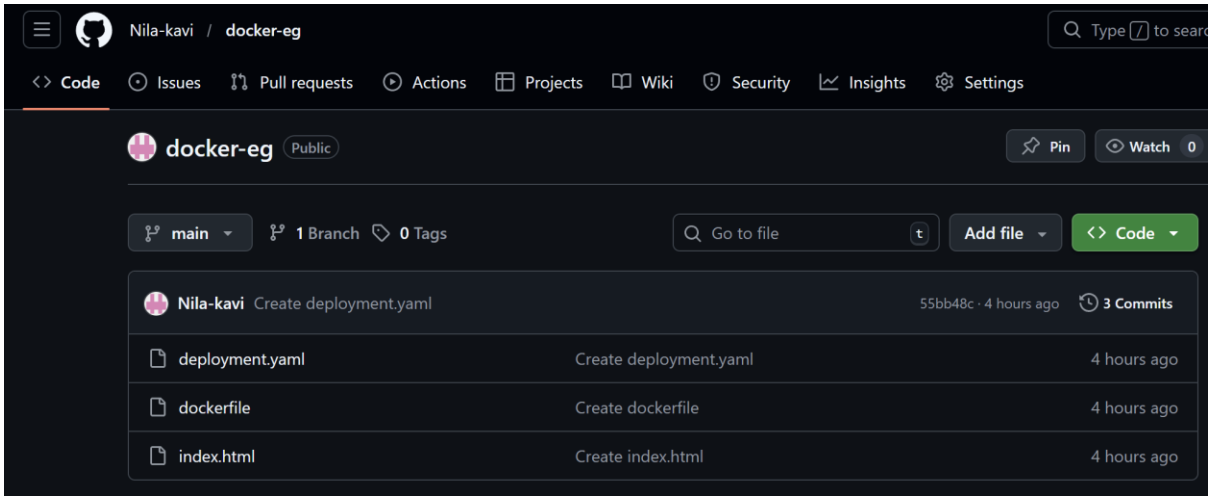
LAB ASSESSMENT-4

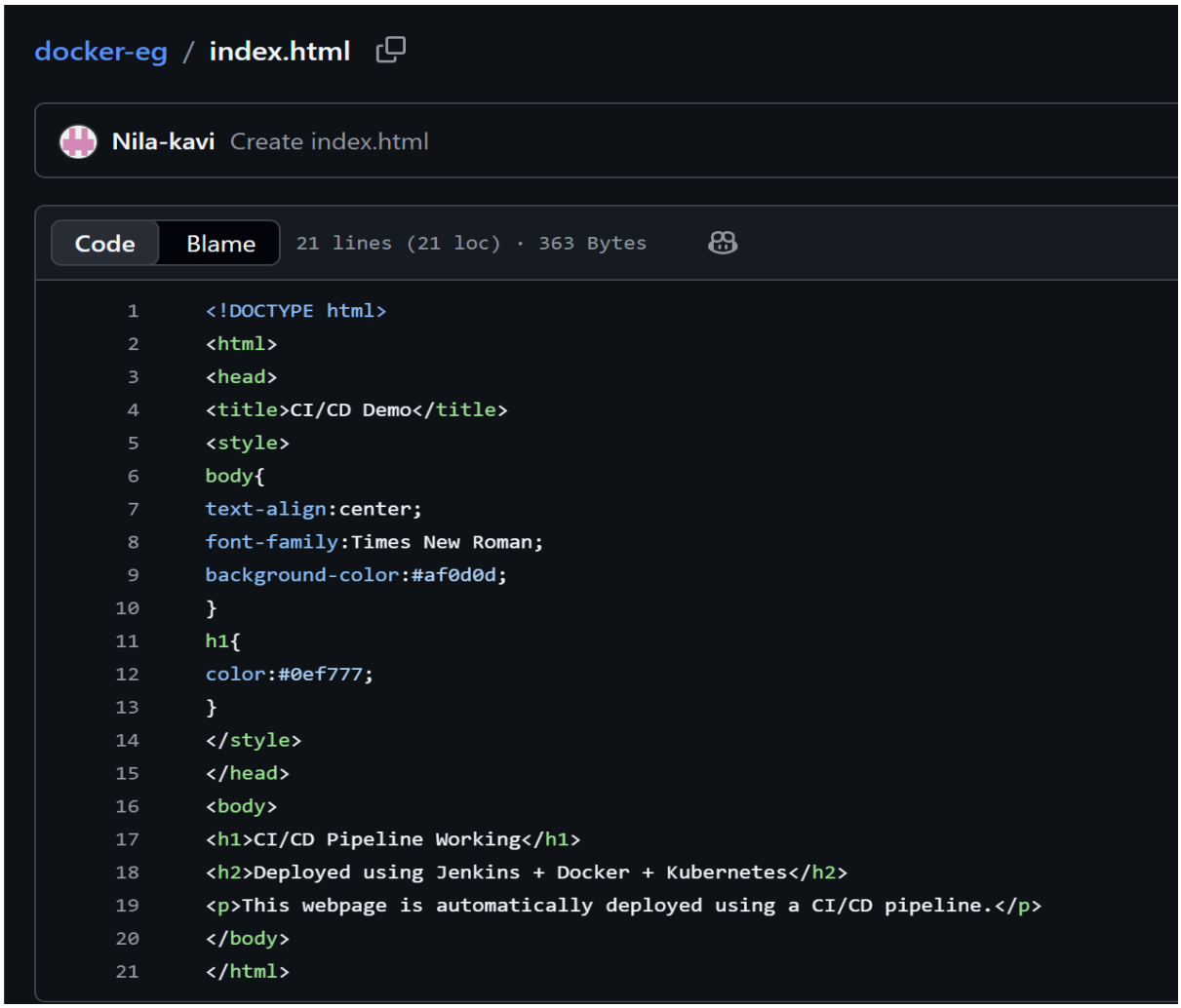
COURSE CODE: ISWE406P

SLOT: L49+L50

FACULTY: SWETA BHATTACHARYA

1. Create a repository in GIT and create the following files:
   - index.html
   - Dockerfile
   - Deployment.yaml






```
<!DOCTYPE html>
<html>
<head>
<title>CI/CD Demo</title>
<style>
body{
text-align:center;
font-family:Times New Roman;
background-color:#af0d0d;
}
h1{
color:#0ef777;
}
</style>
</head>
<body>
<h1>CI/CD Pipeline Working</h1>
<h2>Deployed using Jenkins + Docker + Kubernetes</h2>
<p>This webpage is automatically deployed using a CI/CD pipeline.</p>
</body>
</html>
```

docker-eg / dockerfile

Nila-kavi Create dockerfile

Code Blame 3 lines (3 loc) · 77 Bytes

```
FROM nginx:latest
COPY index.html /usr/share/nginx/html/index.html
EXPOSE 80
```

docker-eg / deployment.yaml

Nila-kavi Update deployment.yaml

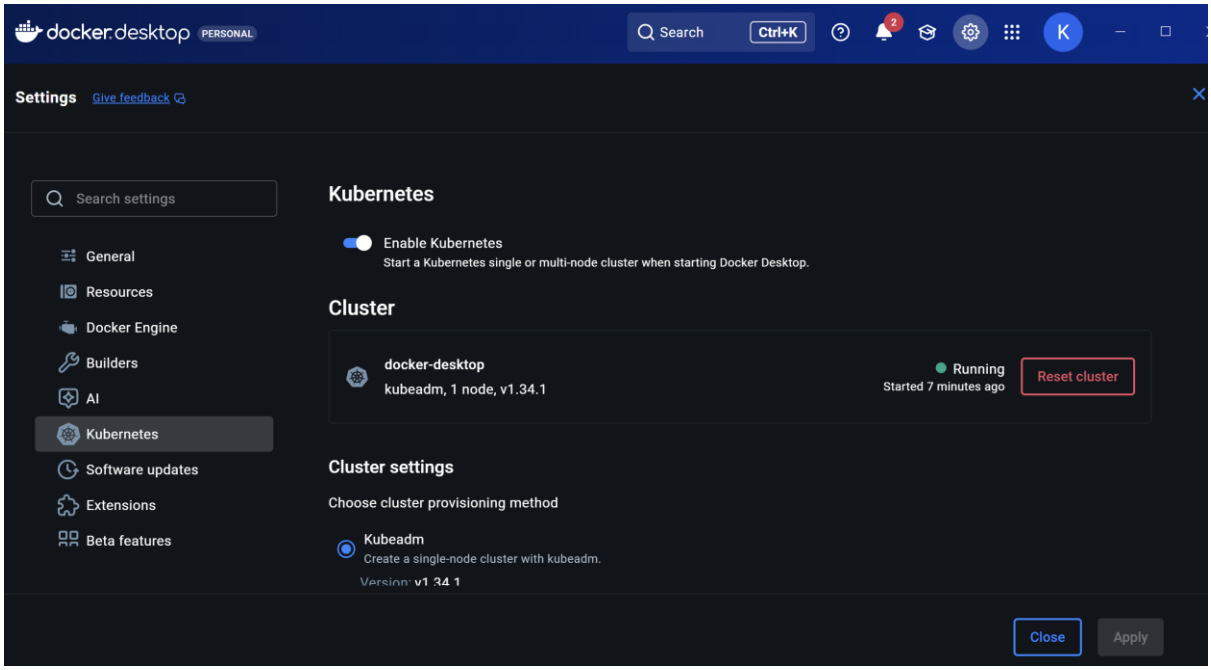
Code Blame 32 lines (32 loc) · 538 Bytes

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: html-demo
spec:
  replicas: 2
  selector:
    matchLabels:
      app: html-demo
  template:
    metadata:
      labels:
        app: html-demo
    spec:
      containers:
        - name: html-demo
          image: kavinila2023/html-demo:latest
          ports:
            - containerPort: 80
---
apiVersion: v1
kind: Service
metadata:
  name: html-demo-service
```

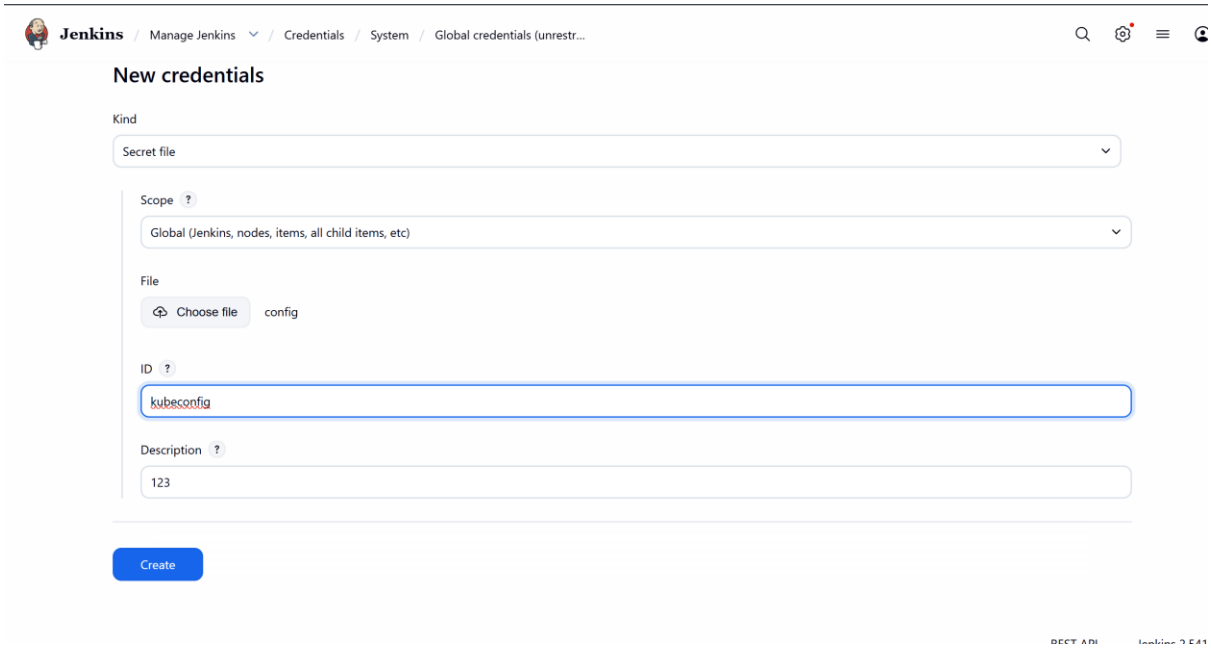

```yaml
            - containerPort: 80
---
apiVersion: v1
kind: Service
metadata:
  name: html-demo-service
```


```yaml
spec:
  type: NodePort
  selector:
    app: html-demo
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30010
```

2. Enable Kubernetes



3. Open Jenkins
   In Jenkins Setting --> Credentials



Jenkins / Manage Jenkins / Credentials / System / Global credentials (unrestr...

Global credentials (unrestricted)

Credentials that should be available irrespective of domain specification to requirements matching.

kavinila/******
fbbe01e8-dc3d-4c47-a23f-7eb1199ae1ea

kavinila2023/****** (1232)
dockerhub - 1232

config (123)
kubeconfig - 123

Create New Item -> Pipeline

Jenkins / dockerhub / Configure

Configure

- General
- Triggers
- Pipeline
- Advanced

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

Script ?

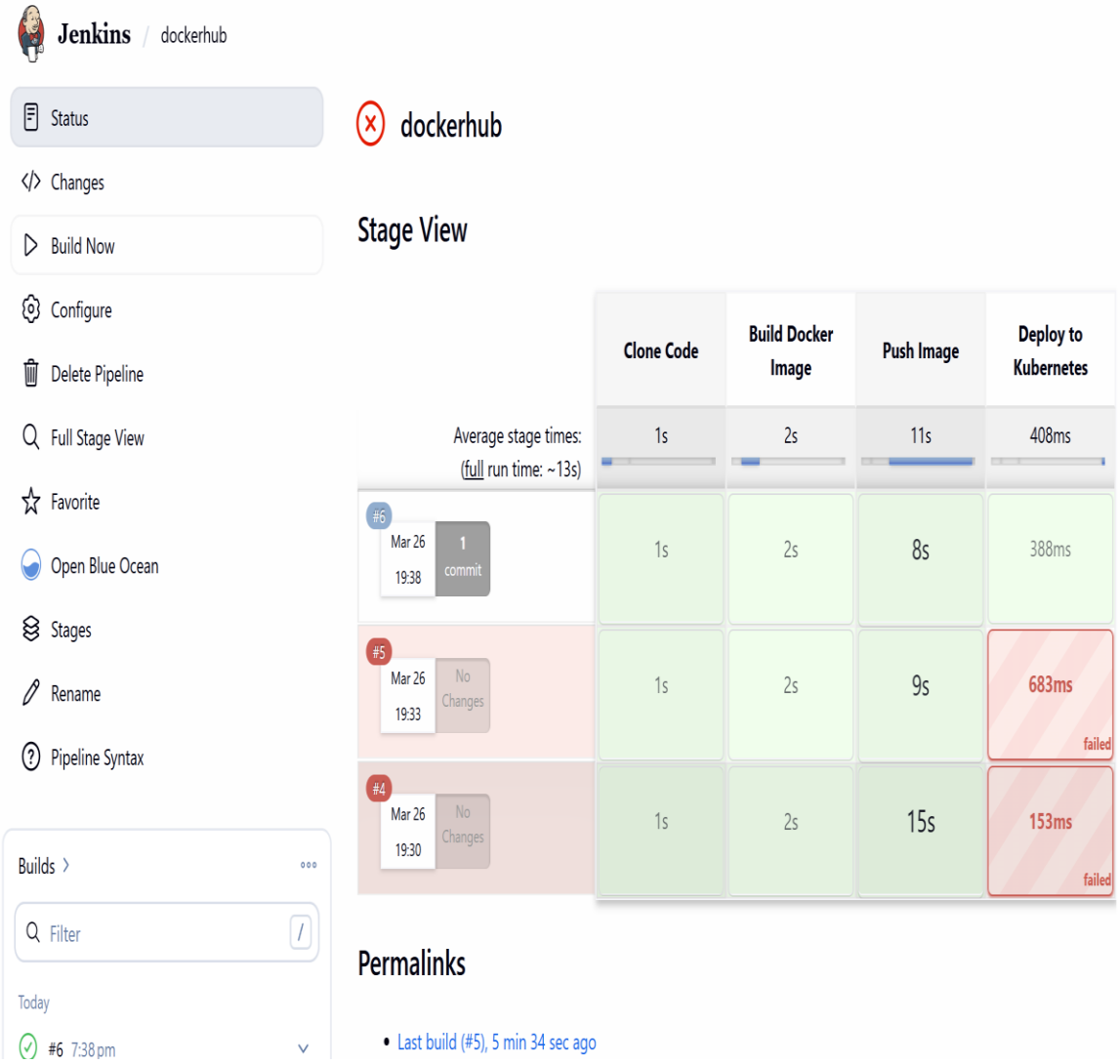
```
pipeline {
    agent any

    environment {
        DOCKER_IMAGE = "kavinila2023/html-demo"
    }

    stages {

        stage('Clone Code') {
            steps {
                git branch: 'main',
                url: 'https://github.com/Nila-kavi/docker-eg.git'
            }
        }
```

Pipeline script

Script ?

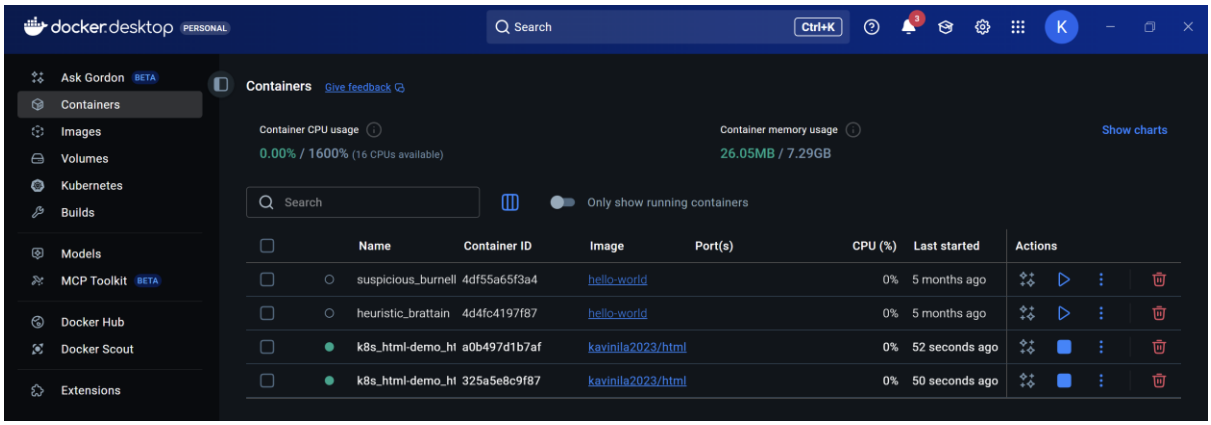
```
bat 'docker build -t %DOCKER_IMAGE% .'
}
}
stage('Push Image') {
steps {
withCredentials([usernamePassword(credentialsId: 'dockerhub',
usernameVariable: 'USER', passwordVariable: 'PASS')]) {
bat 'docker login -u %USER% -p %PASS%'
bat 'docker push %DOCKER_IMAGE%'
}
}
}
stage('Deploy to Kubernetes') {
steps {
withCredentials([file(credentialsId: 'kubeconfig', variable:
```

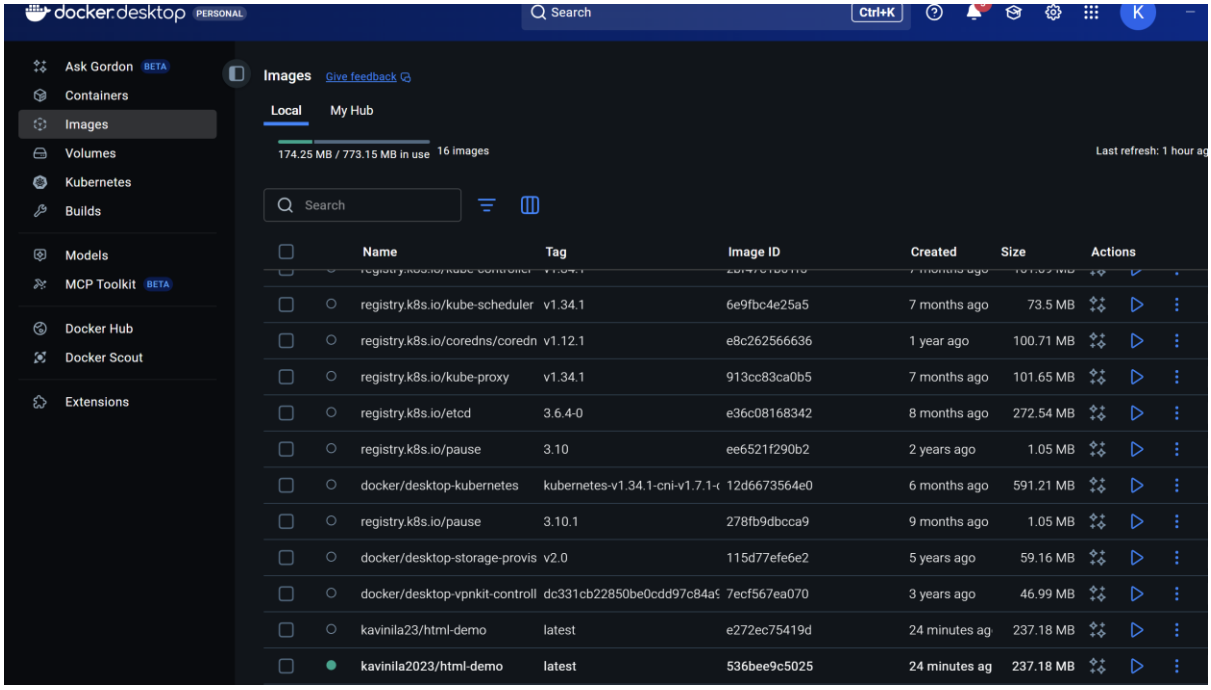
```
steps {
withCredentials([file(credentialsId: 'kubeconfig', variable:
'KUBECONFIG')]) {
bat '''
set KUBECONFIG=%KUBECONFIG%
kubectl apply -f deployment.yaml --validate=false
'''
}
}
}
}
}
```

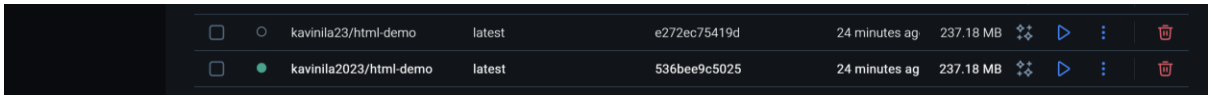


Jenkins / dockerhub / #6

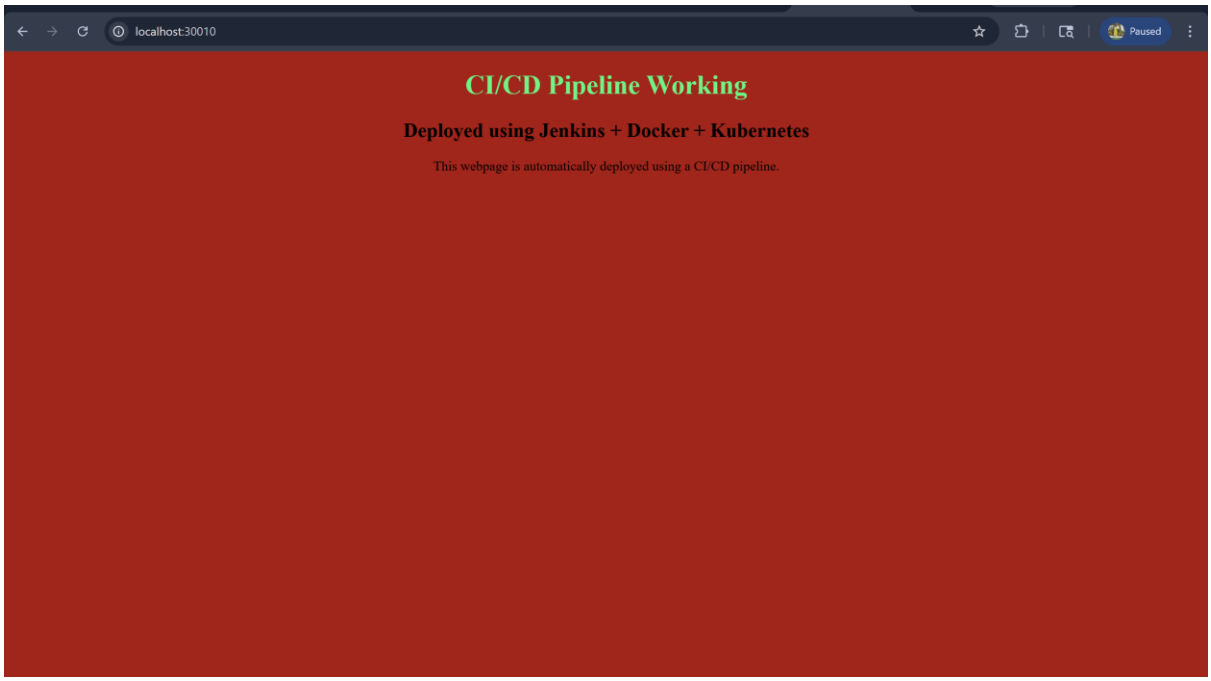
```
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy to Kubernetes)
[Pipeline] withCredentials
Masking supported pattern matches of %KUBECONFIG%
[Pipeline] {
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\dockerhub>set KUBECONFIG=****

C:\ProgramData\Jenkins\.jenkins\workspace\dockerhub>kubectl apply -f deployment.yaml --validate=false
deployment.apps/html-demo created
service/html-demo-service created
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```







Also the HTML page should open in the port mentioned in deployment.yaml file: nodePort: 30010\

So check if the HTML file open in

http://localhost:30010